

Part 1. Device simulation and dataset construction

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We describe the simulated double-quantum-dot devices from which every charge stability diagram in this work is drawn, and the train/test split applied to them. 550 devices are generated from a constant-capacitance model with randomly drawn capacitances, each rendered as a 100 x 100 pixel charge-sensor map over a 2 x 2 mV gate-voltage window. The simulation is noise-free. Devices are split by device identity, not by image, so no device contributes to both sets.

1. Simulated devices

Each device is a double quantum dot with a nearby charge sensor, simulated with the QArray constant-capacitance model. The capacitance matrices are drawn at random from fixed intervals; no draw is filtered, checked or redrawn, so whatever the intervals produce enters the pool. The intervals are chosen so that each dot is driven mainly by its own plunger gate, which is the condition for the two families of transition lines to remain distinguishable.

The sensor response is computed with the noise model set to NoNoise and nothing is added afterwards, so the maps are noise-free. The Coulomb peak width is 0.01 and the electron temperature 1e-05 in simulation units.

Each device is rendered over a 2 x 2 mV window at 100 x 100 pixels. The window is randomly offset per device (offset scale 0.35), so the honeycomb is not registered to the frame and the network cannot learn a fixed position for the lines.

The raw sensor map is dominated by a large, smooth gate-to-sensor cross-talk background, with the charge transitions riding on it as a small modulation. Figure 1 shows the same device before and after that slow background is removed; the honeycomb is visible only in the second. This is the reason the reconstruction problem is not trivial.

2. Ground truth

For every device the simulator also returns the exact charge-transition map: a binary image, one pixel wide on each transition line. Transition-line pixels are 7.1 % of the diagram, which is what makes pixel accuracy useless as a score and motivates the metric of Part 4.

3. Train/test split

THE THING THAT IS SPLIT IS THE DEVICE, NOT THE IMAGE. Capacitance configurations are drawn first and each is given an identity; the identities are split once, with seed 12345, into 500 training and 50 held-out devices. Every image and every measurement budget inherits the identity of the device it came from, so a device's data always lands on one side of the split.

The split is stored with the device pool rather than with a configuration. Every cell of a sweep therefore reads the same stored assignment, and a device cannot be a training device for one measurement budget and a test device for another.

A further 15 % of the TRAINING devices (75 of 500) is set aside before any training as a validation split, leaving 425 to fit the weights. This validation split is where the binarisation threshold of Part 4 is chosen. The 50 held-out devices are never used for any decision.

Figure 2 shows the two levels of the split.

Figures

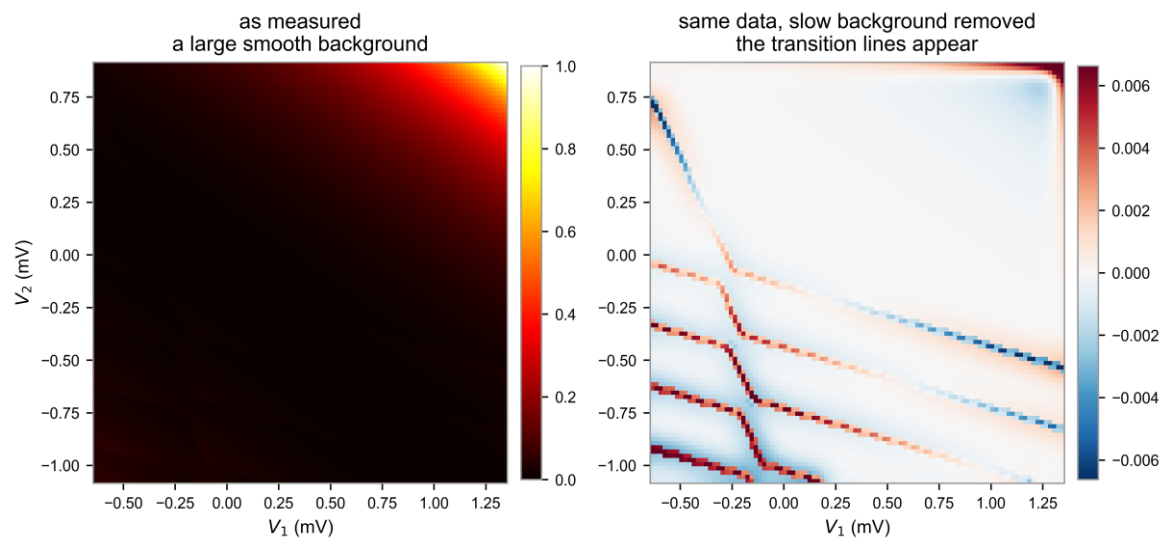


Fig. 1. One simulated device. Left: the charge-sensor map as measured, dominated by a smooth gate cross-talk background. Right: the same data with that slow background differenced away, where the honeycomb of charge transitions becomes visible.

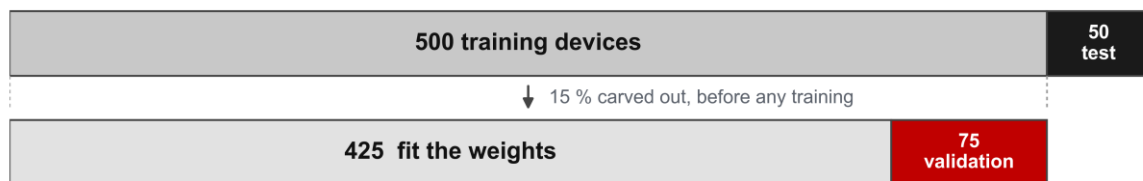


Fig. 2. The train/test split. 550 devices are divided by device identity into 500 training and 50 held-out devices (seed 12345). Inside the training devices, 15 % is carved out as a validation split before any training, leaving 425 devices to fit the weights.